

Intelligent Transportation Scheme for Autonomous Vehicle in Smart Campus

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Abstract— Autonomous Vehicles are being introduced to our society on a daily basis, the whole concept of IoT has an ever growing presence, by combining the two concepts into one “Quasi-Smart City” control and communication scheme, we can get an integrated system whose first design approach is presented in this paper along with the analysis of integration techniques and protocols to develop a smart vehicle inside a state of the art communication environment which is a Smart Campus, where autonomous cars, different sensors, school data, and smart signaling combined, conform a whole new network to offer new services to the academic community such as transporting across the University Campus, smart energy appliances and smart parking among other benefits which are intended to assist general and impaired individuals who are exposed to a very high temperature during the summer time.

Keywords— *Autonomous vehicles, smart campus, control systems, IoT, Kinect, LIDAR, self-driving car.*

I. INTRODUCTION AND JUSTIFICATION

Autonomous systems are being subject to studies from a vast variety of industries and sectors, and there are a lot of different perspectives and possible approaches to follow in order to maintain a general scheme for automation purposes.

Self-driving cars have been a much desired goal to attain for decades and we have come a long way since 2005's DARPA Grand Challenge [1]. Although autonomous navigation is still being developed and tuned to perfection, there already are some commercial solutions out there that offer semi-autonomous driving or assisted driving; such as: Google's and Waymo's self-driving car, TESLA's Autopilot function in all of their cars, Ford, GM, Nissan, Daimler, VW, among others [2, 3].

Nevertheless, all of the aforementioned solutions seek to provide autonomous transportation as a replacement for our existing means of transporting to our home, our workplace or even for travelling through the highway to different cities. Our proposed solution is a novel application of autonomous driving to provide safe, fast and accurate transport to people inside a university campus, between parking lots, buildings and other specific areas of interest within said compound.

It is important to emphasize that we do not mean that this vehicle will travel through the same streets of the campus

through which passenger vehicles transit, but that the autonomous vehicle will navigate within the campus through the same wide roads that pedestrians and bicycles use to travel between buildings. This means that the problem of autonomous navigation that we are solving is different from that faced by companies whose purpose is that their vehicles transit autonomously through city streets and freeways.

There are few developments in Mexico related to autonomous vehicle control using digital image processing and fuzzy control [4]. Other developments take into consideration robot-type vehicles maneuvering in disordered environments with mobile obstacles and using probabilistic algorithms to evade such obstacles [5].

Our approach is based on the utilization of several sensors to map the environment and plan the trajectory of the electric car to navigate it without human intervention and all of this using IoT platforms. The system also incorporates solar energy equipment, composed of photovoltaic panels and battery banks, which is assisted by a PWM charge controller with the primary function of managing the charging and discharging path between the batteries as seen on [6] and [7] which proved to be sustainable and commercially feasible solutions applied to unmanned robotic vehicles, in contrast, our solution is applied to the transportation of students, professors, personnel and visitors inside the academic community.

The project described in this paper is being developed in Mexicali, the capital city of the Mexican state of Baja California and it is planned to be implemented at CETYS University, Mexicali Campus. Being located in one of the hottest cities in Mexico and according with the National Institute of Ecology the city with the highest rate of fatalities caused by heat waves in Mexico [8], in specific, from recollected data of Mexicali's General Hospital during the period from 2006–2010, 76 patients with heat wave symptoms were admitted and 23 of them died [9].

Furthermore, studies have shown that elderly population are among the most vulnerable to heat waves and elevated temperatures [10], given the previously mentioned statistics and the additional factor that within the campus there are elderly people, the transportation within the campus during the summer, becomes a dangerous task that we are seeking to assist with the implementation of an autonomous vehicle.

Finally, in the pursuit of achieving inclusiveness in our campus, this solution will benefit the people with mobility or visual impairment, where there have been several breakthroughs in assistive technologies for use in the classroom as presented on [11-14] but never in terms of secure transportation systems that allow the transit between classes, and to and from the school parking lot, where in several scenarios the health of people on campus could be at risk.

II. PROJECT OBJECTIVES

Our general objective with this research is to develop an intelligent transportation system implemented in an automobile structure capable of transporting an individual from point A to B avoiding obstacles with all of the security measures necessary in order to maintain safety for the passenger.

The specific objectives of this research include but are not limited to the following:

- Develop a 3D vision system capable of detecting the road, potential static and dynamic obstacles (or dangerous conditions), among other variables.
- Design an autonomous navigation system capable of safely traversing a path from point A to B using the 3D vision system data and re-route the vehicle if needed to achieve the desired location.
- Build a sustainable electrical system for the vehicle, with a battery capable of recharging itself through solar energy.
- Design a network infrastructure to suffice the whole communication scheme and fulfill the requirements of a Smart Campus.

It is important to highlight that the scope of this paper is to present our approach to the concept of a Smart Campus and introduce the Intelligent Transportation Scheme as the first system of our Smart Campus initiative. The specific objectives mentioned above are being worked on but at the time of writing this article, they have not been completed in full. Nonetheless, the progress and current status of the autonomous vehicle prototype is presented.

III. AUTONOMOUS CARS & SMART CAMPUS

At this point in time, there is much information related to smart devices. Also, there is a vast quantity of research being conducted which contemplate the actual use and the future of autonomous cars, including different means of transport, a whole different scheme for communications and signaling, among others [15].

After the recent development of IoT applications [16], several engineering efforts have lead towards this scenario, where all real-time data obtained from the transportation systems' sensors becomes available in the cloud for further computing and analysis for different systems to interact with an agent. The agent receives and executes the instructions to travel around the campus in a safe way. In general, the system can be composed of various agents that aim to transport people around the campus and thus form a multi-agent system (MAS) [17].

In order to maintain all this scheme working, a whole range of state of the art technology has to be implemented and integrated to fulfill the whole communication scheme's data exchange [18]. A robust methodology for simulation of IoT before the implementation is presented in [19], in order to validate adding more sensors, devices, or agents to the whole system.

To be considered a smart city several factors need to be considered, but the main ones are: high availability, smart sensor's array, a data exchange communication protocol, security, and compatibility [20]. After implementing this requirement in our network our system can become smart.

This is where a new concept comes to mind: The Smart Campus, where a smart city becomes a network paradigm adopted by a university campus in order to maintain all the data infrastructure and offer different services to the students and to all people on campus. Since our goal is to develop a smart agent inside a smart network, some technologies should be adopted as a standard, for example, the sample rate for sensors, the data communication scheme, how systems interact with each other, how all the microcontrollers behave as master and slave depending of the network itself, etc.

A similar smart car was proposed implementing a microcontroller system and all the serial data communication available for it as well [21], to structure a similar project, the difference is seen in the implementation. In order to satisfy the need of a smart agent in the smart campus' network, not only should an individual scheme be considered, but the whole agent to agent, agent to network and network to agent communication should be taken into account to properly form the network architecture.

Therefore, one of the smart agents we talk about in the smart campus would be the autonomous vehicle. The long term goal is to have a fleet of autonomous vehicles navigating through the campus. Users would call a vehicle to their location and tell it where they want to go by selecting a location in a campus map or via voice commands, all through a mobile app which would have the real time location of every autonomous vehicle in campus. Then the system would send the closest vehicle to the user's location. After a user boards a vehicle and confirm the destination, the autonomous vehicle would start navigating to the selected destination. As part of the smart campus, intelligent recharge stations are being planned to be in key parts of the campus; once a trip is completed, the autonomous vehicle would go to the closest station and go into charging/standby mode, to wait for its next trip.

IV. SYSTEMS AND PROTOTYPE DEVELOPMENT

In order to map the surroundings accurately and to properly move along the different route traced by the system, an efficient mapping algorithm must be implemented.

A. SLAM

The navigation algorithm known as SLAM (Simultaneous Localization and Mapping) can provide autonomy to a robot, but a good response time for mapping and localization

problems have to be considered as well [22], in-line optimization algorithms must be implemented in order to achieve real-time processing speed. Among the developed techniques for this purpose, the scene segmentation technique [23-25] in conjunction with a LIDAR proved that it can be successfully applied to identify objects with a fast response.

In this proposal, several sensors inputs are used: a LIDAR sensor which will produce a 3D point cloud that will be processed along with Kinect Frame data and merge both together into a graphic map that contains depth coordinates and color properties of objects.

A simple approach of a SLAM algorithm was planned for the implementation, see figure 1, odometry readings change according to the vehicle movement, the uncertainty from these lectures is treated by the odometry update algorithm. Along with the odometry, a Kinect sensor is proposed because it is fast enough to collect a depth map from a room full of obstacles, within a frame without any problems and the fact that on outdoors applications it has been proven more accurate at detecting obstacles than a camera pair using specific stereo vision techniques [26]; we use this sensor to obtain a fast landmark recognition and an estimation of its position. In addition to that, by using a LIDAR sensor, we should be able to acquire all the necessary environment measurements.

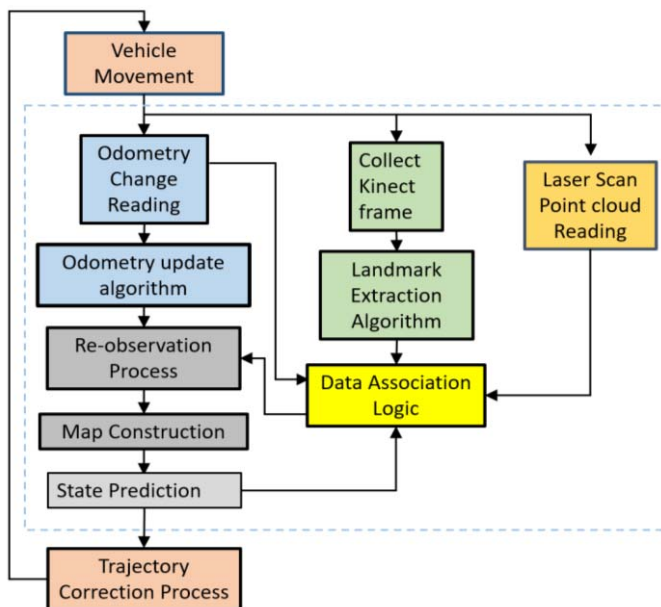


Fig. 1. Diagram of a SLAM Algorithm.

Point cloud readings from Laser sensor and landmark position estimation from Kinect sensor get into an Association Logic Algorithm in order to calculate landmark real position, new landmark position is then used to correct lectures from the odometry algorithm in the re-observation process, finally we obtain a more accurate position from our vehicle, updating the map construction. All this process is to be implemented in real time allowing us to analyze the data and to move faster within the vehicle's desired trajectory.

Based on our previous work [27], where a navigation operation process is described, figure 2 shows the control flow

process algorithm for the autonomous vehicle, after getting current location, the system calculates an initial trajectory and the vehicle will start to move. IR and Ultrasonic sensors are implemented to detect obstacles, if an obstacle is detected the robot vehicle reduces speed or stops to avoid collision. The map construction process intended for scanning the environment and analyzing data is performed during all the trajectory in order to control correct navigation through map corrections based on the obstacles' position. A GPS is available to give information about vehicle's location and to correct initial trajectory or re-route if necessary. The sensors described below should behave synchronously in order to maintain consistency in the whole scheme and for the system to choose the best route to take, and how to re-route correctly taking into account all the obstacles in the vehicle's line of vision.

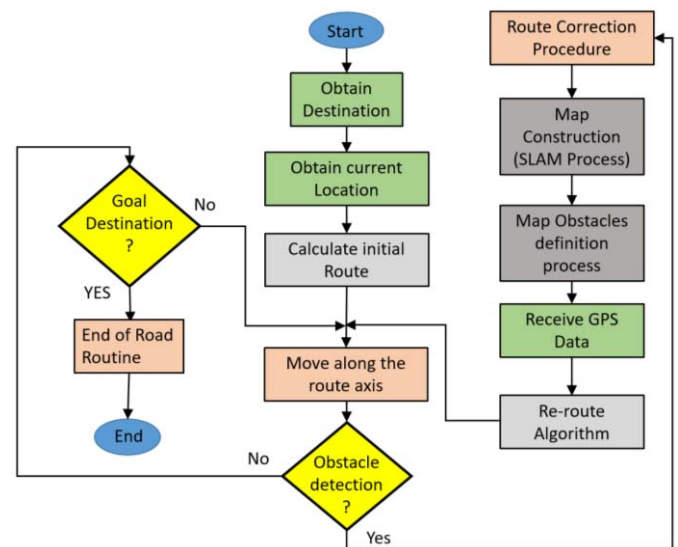


Fig. 2. Control Flow Process for the autonomous vehicle.

B. Python-Flask Server

The core of the control involving communication and signaling through several peripherals is based in a server and client model, where the server listens to different events, and all the sensors connected in the systems submit through channels the data needed in order to maintain the feedback path. By building this kind of schema, all the subsystems remain modular, thus error detection, debugging and system testing become easier and also coding becomes more readable and maintainable in the long run for production. The implementation of the proposed system is based in Python programming language, Flask microframework, uWSGI application container and a high performance REST application programming interface (API) that allows to manage large quantities of data easily in a similar manner as proposed by [28-30].

C. Sensors

Most of the sensors used in this research are meant to measure distance from different perspectives and different transduction principles, and by doing so we ensure the whole

range of vision and perspective therefore, all the “blind spots” should be covered with this arrangement.

1) LIDAR

A Velodyne VLP-16 LIDAR is our main sensor and its use is justified due to the fact that it's capable of delivering the most accurate real-time 3D data that consists in the creation of a full 360-degree environmental point cloud view.

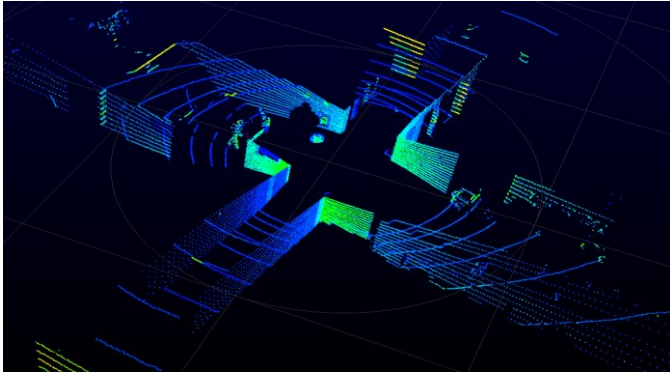


Fig. 3. Real time LIDAR Data Representation at CETYS University.

LIDARs, carry out sensing by using their own emitted light and therefore they are not sensitive to environmental illumination. Road detection systems that rely on this type of sensor can then, in principle, provide the same level of accuracy across the full spectrum of light conditions experienced in daily driving, therefore they are particularly suitable for achieving higher levels of driving automation [31]. Currently we can visualize the LIDAR sensor data as seen in Fig. 3, and we are in the process of transforming and analyzing the data in order to map the autonomous vehicle surroundings and implement it for the navigation task.

2) Kinect Sensor

The Kinect offers a wide variety of sensors, such as an infrared emitter, an infrared depth sensor, a RGB camera that stores three channel data in a 1280x960 resolution, a microphone array which contains four microphones for capturing sound as well as finding the location of the sound source and an infrared camera as well. These sensors offer a huge advantage for systems that depend on the perception of depth to classify obstacles, such as this project.

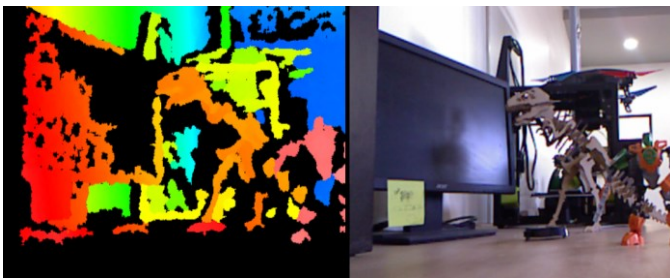


Fig. 4. Depth sense using Kinect sensor.

Experimentation with the Kinect sensor consisted on a program that shows the Kinect's regular camera output, while also storing the infrared depth sensor output, so that when the user points the cursor at a certain pixel in the image, the depth

from the depth array is retrieved and shown on the screen. See Fig. 4.

The official Kinect specification states that the depth range of operation is between 0.8 meters and 4 meters, however on the performed tests, measurements between 0.5 meters and 6 meters were possible, with less than 0.05 meters of error, and results were available in less than 60 milliseconds.

D. Data Acquisition System

The National Instruments roboRIO advanced robotics controller was selected due to its complete and robust processing capabilities which include built-in ports for integrated circuits (I2C), serial peripheral interfaces (SPI), RS232, USB, Ethernet port, relays along with digital and analog inputs and outputs, to seamlessly connect and control sensors and actuators.

This roboRIO serves as the main controller of low level processes and signals, such as controlling motors and reading encoders, IR and ultrasonic sensors, IMU, among others. Finally, it will serve as a bridge to connect to the main PC server aboard the vehicle which will run the high level processes and communications, such as interpreting LIDAR data, 3D mapping, trajectory calculation, and machine learning, among others.

E. Mechanical Design and Kinematic Model

Our mechanical design takes into account the vehicle's chassis, gearboxes, motors, wheels, sensors, batteries, computer, Ethernet switch, cable management, and airflow. The whole autonomous system (vision and control) is mounted on an aluminum chassis with 4" Omni directional wheels in order to satisfy the requirements of forces applied to the base of the vehicle, so all forces can be dissipated and damped. This type of drive allows linear and rotary motion through space which provides the specific advantage of maneuvering in all directions without steering the wheels, which allow for more accurate control and precise movements. See Fig. 5.



Fig. 5. Autonomous vehicle scale prototype.

An electrical DC motor with gearbox are installed on each of the four mecanum wheels of our design, giving our vehicle omnidirectional movement. A kinematic analysis and the necessary transform equations were developed to simulate and control the position and orientation of vehicle in the three dimensional space.

What is planned for the near future in terms of mechanical design will include all the mentioned components along with a vehicle bodywork for esthetics, aerodynamics and external sensors position.

For the adequate control of the vehicle the following inverse kinematic solution from [32] is presented in (1):

$$\begin{bmatrix} \dot{x}_r \\ \dot{y}_r \\ \dot{\theta}_r \end{bmatrix} = \frac{R}{4} \begin{bmatrix} -1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ \frac{1}{\alpha} + b & \frac{-1}{\alpha} + b & \frac{-1}{\alpha} + b & \frac{1}{\alpha} + b \end{bmatrix} \chi \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} \quad (1)$$

where $\dot{\theta}_i$ ($i = 1, 2, 3, 4$) is the angular velocity of each wheel.

The algorithm proposed for the vehicle motion is intended to be a robust trajectory tracking controller like [32], where a high order sliding mode control law was derived and adaptive laws were designed for the auto adjustment to avoid uncertainties and get stability. Also a manual control was designed in this first approach, the programmatic solution for manual control and drive testing via joystick controller is given by the following equations:

$$V_d = \sqrt{J_{left,x}^2 + J_{left,y}^2} \quad (2)$$

The resultant velocity magnitude and direction in the fixed coordinate axis (x, y, z) can be achieved with a joystick movement

$J_{left,x}$ and $J_{left,y}$ variables represent proportional read values from joystick position for axis x and y , thus V_d magnitude is the relation between the angle formed by the 2 axis (x, y) of the controller's stick to drive the mecanum wheels, see (2).

$$V_\theta = -J_{right,x} \quad (3)$$

Changing direction speed is defined by the position of the right stick of the controller $J_{right,x}$, regarding the rotation of the vehicle itself (3).

$$\theta_d = \arctan(-J_{left,x}, J_{left,y}) \quad (4)$$

Resultant velocity direction of "Vd" refers to the angle described by the joystick as seen in (4).

$$\begin{aligned} V_{FL} &= V_d \sin\left(\theta_d + \frac{\pi}{4}\right) + V_\theta \\ V_{FR} &= V_d \cos\left(\theta_d + \frac{\pi}{4}\right) - V_\theta \\ V_{BL} &= V_d \cos\left(\theta_d + \frac{\pi}{4}\right) + V_\theta \\ V_{BR} &= V_d \sin\left(\theta_d + \frac{\pi}{4}\right) - V_\theta \end{aligned} \quad (5)$$

Where VFL is the velocity in front left wheel, VFR is the velocity in front right wheel, VBL is the velocity in back left wheel, VBR is the velocity in back right wheel.

These velocities (5) control each wheel of the mecanum system regarding the position and the angle desired by the movement of the controller stick by implementing a shift of 45° angle (inclination of the rollers of the mecanum wheels).

F. Inertial Measurement Unit

The IMU provides information about position and orientation which will be used to obtain said data of the vehicle in real time without relying specifically in the GPS.

The task of obtaining the data from the sensor is fairly straightforward however, to compute the angles and the correct position of the system, some calculations need to be done to the obtained data, as shown in figure 6.

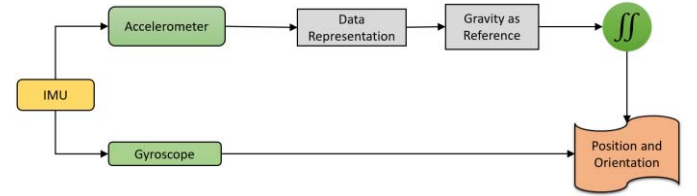


Fig. 6. IMU Data Representation.

We used a discontinuous point by point smoothness function to approximate the graph of different measured points in 3D space by the IMU sensor, thus generating a better understanding of the dynamic behavior of our system. By using a spline smoothness function in MATLAB, we could interpolate a curve into a set of points measured and recorded by the IMU system. By doing so, we could estimate how the whole vehicle system is moving and how it should behave in the following re-routings.

V. ARCHITECTURE

Since Supervisory Control and Data Acquisition have become more popular and gained an important role in the industry, this project's architecture is made to resemble that of an Industrial and Automation Control System with an important connection with SCADA systems.

According to Geng in the SCADA systems and IoT handbook [33], IoT is just an extension and another tool for SCADA systems to take advantage from. The complete scheme of the project was developed around the concept of making the whole system modular, and keeping all subsystems interconnected, controlled by a main central processing unit, and also with a variety of pathways for data exchange, communication with each other, and with the end user.

The automation goal is achieved since the communication is established from the sensors to the user interface, and from the user interface to the actuators, closing the loop for a complete automation and control scheme with built-in tools for data analysis and statistical representation of efficiency per tuning in the system contemplating system's fluctuation.

Based on the automation levels describing the SCADA system in general exposed by L. Rosa et al [34], our system is structured following the five distinct principles or levels for automation, according to level 0 our system has several sensors and actuators for measurement, control and user interaction actions. Level 1 is accomplished by implementing programmable logic unit and remote terminal units following our decentralized computational scheme for data acquisition and the communication protocols for data exchange to our central processing unit. Level 2 is completed using several web services as HMI in order to enable the user with a whole range of possibilities for interactions with the system. Level 3 and 4 are accomplished with both the computational level available in the user end and the system's central processing unit, completing all the planning for the system to work properly. Our proposed architecture scheme is shown in figure 7.

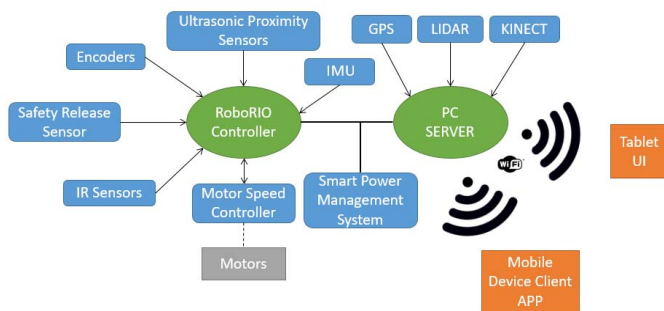


Fig. 7. General Interconnection Scheme for the Autonomous Vehicle System.

VI. CONCLUSIONS AND FUTURE WORK

After testing a variety of complex sensors (such as the LIDAR and Kinect IR Camera) and actuators available and calibrating the whole range of set points estimated for the different sensors, it's understood that the complexity of the system described, is very high, and the behavior expressed by the whole system taking in consideration all the compensation factors, mechanical errors and discontinuities, control instability, and error in calibration could make our system prone to small measurement errors which we have already spotted and are working to correct.

Nevertheless, at this stage of our research, the event of attaining the goals initially proposed has become plausible. Our sensors' and actuators' analysis, the implementation of SLAM algorithms, along with the design of our own architecture and mapping scheme, makes this research very promising and possible to integrate seamlessly in order to create an autonomous agent (vehicle) capable of safely performing the task of navigation described.

As stated before, the purpose of this paper was to establish and present a novel approach to the concept of a Smart Campus, and to introduce the Intelligent Transportation Scheme as the first system of the Smart Campus initiative.

Future work involves finishing the prototype and performing autonomous driving tests at CETYS University,

Mexicali Campus, complete design and experimentation results will be published in a full journal paper.

ACKNOWLEDGMENT

The authors would like to thank Center of Innovation and Design (CEID) of CETYS University for all facilities to perform the research and for providing the necessary equipment to develop this project.

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